A Few Notes on Vector Calculus

David Meyer

dmm@1-4-5.net

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1 Introduction

TBD

2 Notation

Some authors use bold to denote a vector. Adding a "hat" denotes a unit vector; for example $\hat{\mathbf{i}}$ denotes the unit vector in the x direction. Using these definitions the parametric form of a three dimensional vector $\mathbf{r}(t)$ would be

$$\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}} + k(t)\,\hat{\mathbf{k}}$$

On the other hand, some authors use "over arrow"

$$\vec{r}(t) = g(t)\,\vec{i} + h(t)\,\vec{j} + k(t)\,\vec{k}$$

I've also seen

$$\vec{r}(t) = g(t)\,\hat{i} + h(t)\,\hat{j} + k(t)\,\hat{k}$$

In this document I'll use these forms interchangeably.

We will also use $\dot{\mathbf{r}}$, $\dfrac{d\mathbf{r}}{dt}$ and $\mathbf{r}'(t)$ interchangeably.

2.1 Vector Fields

3 Arc Length

Consider a segment of a parametric curve $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}} + k(t)\,\hat{\mathbf{k}}$ between two points $P = \mathbf{r}(t)$ and $Q = \mathbf{r}(t + \Delta t)$, as shown in Figure 1.

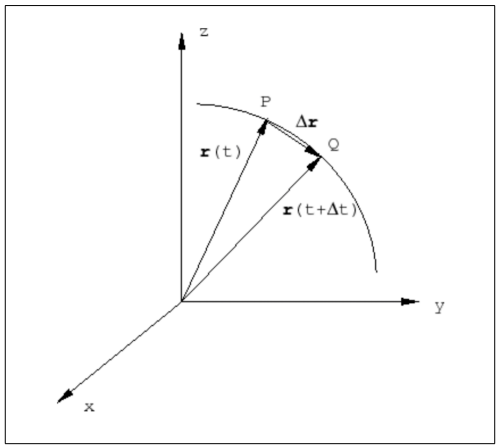

Figure 1: $\Delta\mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$

What is the length of the segment of the curve between P and Q? This length is called the arc length and is denoted by Δs. Δs can be approximated by the chord length $\|\Delta\mathbf{r}\|$. Then we can compute the infinitesimal arc length ds as follows:

$$
\begin{aligned}
\Delta s &\approx \|\Delta\mathbf{r}\| && \text{\# approximate } \Delta s \text{ by } \|\Delta\mathbf{r}\| \\
&= \|\mathbf{r}(t + \Delta t) - \mathbf{r}(t)\| && \text{\# } \Delta\mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t) \\
&= \|\mathbf{r}(t + \Delta t) - \mathbf{r}(t)\| \cdot \frac{\Delta t}{\Delta t} && \text{\# multiply by } 1 = \frac{\Delta t}{\Delta t} \\
&= \left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# assume } \Delta t > 0 \\
\Rightarrow\ & \Delta s \approx \left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# this is our approximation} \\
\Rightarrow\ & \lim_{\Delta t \to 0} \Delta s = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# is } = \text{ valid here } (\approx \to = \text{ in the limit})? \\
\Rightarrow\ & ds = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# } \Delta t \to 0 \Rightarrow \Delta s \to 0 \Rightarrow \lim_{\Delta t \to 0} \Delta s = ds \\
\Rightarrow\ & ds = \lim_{\Delta t \to 0} \left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# product rule for limits [1]} \\
\Rightarrow\ & ds = \left\| \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# for a normed vector space } X \ \lim \|x_n\| = \|\lim x_n\| \\
\Rightarrow\ & ds = \left\| \frac{d\mathbf{r}}{dt} \right\| \cdot dt && \text{\# } \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{d\mathbf{r}}{dt},\ \lim_{\Delta t \to 0} \Delta t = dt
\end{aligned}
$$

Interestingly we can also see this using the Pythagorean theorem. Specifically, if we move an infinitesimal distance in the x and y directions we get a triangle in the x-y plane who's sides have length dx and dy and who's hypotenuse is ds. This scenario is shown for some curve $f(x)$ in Figure 2.

In the two dimensional case where the parameterization is $x = g(t)$ and $y = h(t)$ we have $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}}$. Further, since derivative is a linear operator [3] we know that

$$\frac{d\mathbf{r}}{dt} = \frac{dg}{dt}\,\hat{\mathbf{i}} + \frac{dh}{dt}\,\hat{\mathbf{j}} \tag{1}$$

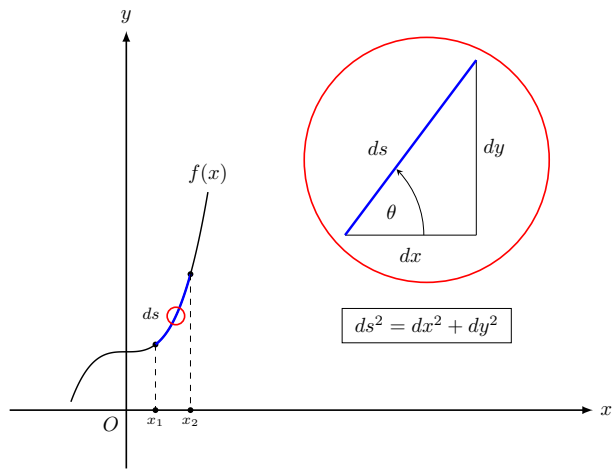

Figure 2: $f(x)$, ds and the Pythagorean Theorem

So in the source domain the Pythagorean theorem tells us that $ds = \sqrt{dx^2 + dy^2}$. Using this expression for ds we can see that

$$\begin{aligned}
ds &= \sqrt{dx^2 + dy^2} && \text{\# Figure 2} \\
&= \sqrt{dg^2 + dh^2} && \text{\# switch to parametric form: } x = g(t) \text{ and } y = h(t) \\
&= \sqrt{dg^2 + dh^2} \cdot \frac{dt}{dt} && \text{\# multiply by } 1 = \frac{dt}{dt},\ dt > 0 \\
&= \sqrt{dg^2 + dh^2} \cdot \left[\sqrt{\frac{1}{dt^2}} \cdot dt\right] && \text{\# } \frac{x}{x} = \sqrt{\left(\frac{1}{x}\right)^2} \cdot x = \sqrt{\frac{1}{x^2}} \cdot x,\ x > 0 \\
&= \sqrt{\left(dg^2 + dh^2\right) \cdot \left(\frac{1}{dt^2}\right)} \cdot dt && \text{\# simplify: } \sqrt{x} \cdot \sqrt{\frac{1}{y}} = \sqrt{x \cdot \frac{1}{y}} \\
&= \sqrt{\left(\frac{dg}{dt}\right)^2 + \left(\frac{dh}{dt}\right)^2} \cdot dt && \text{\# simplify: } \left(x^2 + y^2\right) \cdot \left(\frac{1}{z^2}\right) = \frac{x^2}{z^2} + \frac{y^2}{z^2} = \left(\frac{x}{z}\right)^2 + \left(\frac{y}{z}\right)^2 \\
&= \sqrt{g'(t)^2 + h'(t)^2} \cdot dt && \text{\# Leibniz's notation} \rightarrow \text{Lagrange's notation: } \frac{df}{dt} = f'(t) \\
&= \left\|\frac{d\mathbf{r}}{dt}\right\| \cdot dt && \text{\# Equation (1) and the definition of } \|\cdot\|
\end{aligned}$$

In any event, now we see that

$$\frac{ds}{dt} = \left\| \frac{d\mathbf{r}}{dt} \right\| \tag{2}$$

and so

$$\frac{ds}{dt} = \|\dot{\mathbf{r}}\| \tag{3}$$

Since $\frac{ds}{dt} = \left\| \frac{d\mathbf{r}}{dt} \right\| = \|\dot{\mathbf{r}}\|$ we also know that the unit tangent vector $\mathbf{T} = \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} = \frac{d\mathbf{r}}{ds}$ [2]. One way to see this is to notice that

$$\begin{aligned}
\mathbf{T} &= \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} && \text{\# definition of the unit tangent vector } \mathbf{T} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\|\dot{\mathbf{r}}\|} && \text{\# definition of } \dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\left(\frac{ds}{dt}\right)} && \text{\# Equation (3): } \|\dot{\mathbf{r}}\| = \frac{ds}{dt} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\left(\frac{ds}{dt}\right)} \cdot \frac{dt}{dt} && \text{\# multiply by } 1 = \frac{dt}{dt} \\
&= \frac{d\mathbf{r}}{ds} && \text{\# simplify}
\end{aligned}$$

4 Conclusions

Acknowledgements

LaTeX Source

References

[1] Alex Svirin. Properties of Limits. `https://math24.net/properties-limits.html`. [Online; accessed 14-November-2021].

[2] Gregory Hartman. Unit Tangent and Normal Vectors. `https://sites.und.edu/timothy.prescott/apex/web/apex.Ch12.S4.html`, Nov 2021. [Online; accessed 08-March-2022].

[3] Wikipedia contributors. Linearity of Differentiation — Wikipedia, The Free Encyclopedia. `https://en.wikipedia.org/w/index.php?title=Linearity_of_differentiation&oldid=1026177205`, 2021. [Online; accessed 1-September-2021].